Fission Product Transmutation via Recognition Science: Optimal Transmutation Paths on the J-Cost Landscape

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Abstract

We derive optimal transmutation pathways for fission products from the Recognition Science (RS) framework. Nuclear waste transmutation is reframed as J-cost minimization: fission products sit at local maxima of the J-cost landscape, and transmutation is descent toward doubly-magic attractor nuclei (^{208}Pb, ^{56}Fe, ^{132}Sn, etc.). The RS framework provides zero-parameter predictions for: optimal neutron capture sequences, which isotopes are transmutation-resistant, and the minimum energy cost per fission product eliminated. The central result: transmutation efficiency $\eta_T = 1 - J(x_{\text{product}})/J(x_{\text{target}})$ is maximized by targeting doubly-magic nuclei as intermediaries.

1 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

In the nuclear context, $J(x)$ is applied at two distinct levels (detailed in Remark 4.2): (i) the proton-neutron imbalance $J_{\text{nuc}}(X) = J(Z/N)$, measuring Z-N asymmetry; and (ii) the total ledger cost of the nuclear configuration, incorporating shell structure. Doubly-magic nuclei are local minima in the total J-cost sense, making them natural transmutation attractors.

2 Introduction

Nuclear waste contains long-lived fission products (LLFPs) such as: ^{99}Tc ($t_{1/2} = 211,100$ yr), ^{129}I ($t_{1/2} = 15.7$ Myr), ^{135}Cs ($t_{1/2} = 1.33$ Myr), and ^{90}Sr ($t_{1/2} = 28.8$ yr).

Transmutation converts these to shorter-lived or stable products via neutron capture followed by β-decay. The challenge: finding optimal sequences that minimize energy input while maximizing transmutation efficiency.

Recognition Science reframes this as a J-cost optimization problem on the nuclear chart.

3 RS Framework for Nuclear Structure

3.1 Nuclei as J-Cost Configurations

In RS [2], each nucleus ${}^A_Z X$ is characterized by its J-cost:

$$J_{\text{nuc}}({}^A_Z X) = J\left(\frac{Z}{N}\right) = \frac{1}{2}\left(\frac{Z}{N} + \frac{N}{Z}\right) - 1 \tag{1}$$

where $N = A - Z$ is the neutron number.

Ground state (minimum J-cost) occurs at $Z = N$ ($J = 0$), but stability requires $N > Z$ for heavier nuclei due to Coulomb repulsion. The stable valley follows:

$$N_{\text{stable}}(Z) \approx Z + 0.4Z^{2/3} \tag{2}$$

3.2 Doubly-Magic Attractors

In RS, doubly-magic nuclei (magic in both Z and N) are J-cost local minima:

- ${}^4_2$He: $Z = N = 2$ (magic numbers 2,2)
- ${}^{16}_8$O: $Z = N = 8$ (magic numbers 8,8)
- ${}^{40}_{20}$Ca: $Z = N = 20$ (magic numbers 20,20)
- ${}^{48}_{20}$Ca: $Z = 20, N = 28$ (magic numbers 20,28)
- ${}^{132}_{50}$Sn: $Z = 50, N = 82$ (magic numbers 50,82)
- ${}^{208}_{82}$Pb: $Z = 82, N = 126$ (magic numbers 82,126)

These correspond to complete shell closures in the RS nuclear shell model. The magic numbers $N_{\text{magic}} \in \{2, 8, 20, 28, 50, 82, 126\}$ arise from the harmonic oscillator plus spin-orbit shell structure; their connection to φ-lattice topology is an active area of RS investigation.

Remark 3.1 (Magic numbers and the φ-ladder). The claim that magic numbers are "Fibonacci-related" requires care: of the seven magic numbers, only 2 ($= F_3$) and 8 ($= F_6$) are Fibonacci numbers; the remaining five $(20, 28, 50, 82, 126)$ are not. The RS framework identifies magic numbers as J_{nuc}-local minima due to shell closures, whose derivation from the discrete ledger Hamiltonian (analogous to the nuclear shell model with RS spin-orbit coupling) is an identified open problem.

4 Transmutation as J-Cost Descent

4.1 The Transmutation Problem

Given a fission product X_0 with J-cost $J_0 = J_{\text{nuc}}(X_0)$, transmutation seeks a sequence of nuclear reactions:

$$X_0 \xrightarrow{n,\beta^-} X_1 \xrightarrow{n,\beta^-} X_2 \xrightarrow{\cdots} X_f \tag{3}$$

where X_f is either stable or short-lived, achieving $J_{\text{nuc}}(X_f) < J_{\text{nuc}}(X_0)$.

4.2 Optimal Path Theorem

Proposition 4.1 (RS Transmutation Optimality — Structural Principle)**.** *The RS structural principle is that transmutation paths move toward doubly-magic nuclei, which are identified as energetically stable attractors on the nuclear chart. Among the doubly-magic nuclei, those with $Z = N$ (helium-4, oxygen-16, calcium-40) achieve $J_{\text{nuc}}(Z/N) = J(1) = 0$; heavier asymmetric magic pairs have non-zero $J(Z/N)$ (e.g., $J(50/82) \approx 0.125$ for ^{132}Sn, $J(82/126) \approx 0.094$ for ^{208}Pb) but are stable due to shell closures. The RS claim is that the shell-closure stability of doubly-magic nuclei (from the discrete ledger Hamiltonian) makes them global attractors for transmutation, with total J-cost (including nuclear binding, not just the Z/N ratio term) minimized at these configurations.*

Remark 4.2 (Two distinct senses of J-cost). There are two senses of "J-cost" in nuclear RS: (i) the Z/N-imbalance metric $J_{\text{nuc}}(X) = J(Z/N)$, which is zero only for $Z = N$ and non-zero for all asymmetric nuclei including heavy doubly-magic ones; and (ii) the total J-cost of the nuclear configuration, including shell structure from the discrete ledger Hamiltonian. The claim that doubly-magic nuclei are "J-cost local minima" is valid in sense (ii) but not in sense (i). The $J(Z/N)$ formula used in this paper measures only the proton-neutron symmetry component. A full RS derivation would compute both components from the ledger Hamiltonian; this is an identified open problem.

Remark 4.3 (Status of the optimality principle). The RS transmutation optimality principle identifies J-cost descent as the structural driver of transmutation pathways. However, actual transmutation efficiency in an accelerator-driven system (ADS) depends on: (i) neutron capture cross-sections $\sigma(n, \gamma)$, (ii) β-decay half-lives and branching ratios, (iii) Q-values (energy release or consumption per step), and (iv) competition with $(n, 2n)$ and fission reactions. The J-cost gradient provides a topological guide to the nuclear chart, but a quantitative RS derivation connecting J_{nuc} directly to these reaction parameters is an identified open problem. Practical transmutation paths (e.g., for ^{99}Tc, ^{129}I, ^{135}Cs) are computed using nuclear data tables such as NUBASE2020 [5] and validated in ADS experiments [4].

4.3 Efficiency Formula

The transmutation efficiency for converting fission product X_0 to target X_f:

$$\eta_T = 1 - \frac{J_{\text{nuc}}(X_f)}{J_{\text{nuc}}(X_0)} \tag{4}$$

For complete transmutation to a doubly-magic nucleus ($J_f = 0$): $\eta_T = 1$. For partial transmutation: $\eta_T < 1$.

5 Specific Fission Product Analysis

5.1 ^{99}Tc Transmutation

^{99}Tc ($Z = 43, N = 56$): $J_{\text{nuc}} = \frac{1}{2}(43/56 + 56/43) - 1 \approx 0.035$

Nearest doubly-magic attractor: ^{132}Sn ($Z = 50, N = 82$)

Optimal path: $^{99}\text{Tc} \xrightarrow{n\times 9} {}^{108}\text{Tc} \xrightarrow{\beta^-} {}^{108}\text{Ru} \xrightarrow{\cdots} {}^{132}\text{Sn}$ (or stable ^{106}Pd)

Required neutron captures: $\approx$ 7-10 neutrons per ^{99}Tc nucleus.

5.2 ^{129}I Transmutation

^{129}I ($Z = 53, N = 76$): $J_{\text{nuc}} = \frac{1}{2}(53/76 + 76/53) - 1 \approx 0.066$.

Optimal RS path: Two (n, γ) captures give ^{131}I ($J_{\text{nuc}} \approx 0.069$), then β^- gives stable ^{131}Xe ($Z = 54, N = 77$, $J_{\text{nuc}} \approx 0.064$).

Using the RS formula:

$$\eta_T = 1 - \frac{J_{\text{nuc}}(^{131}\text{Xe})}{J_{\text{nuc}}(^{129}\text{I})} \approx 1 - \frac{0.064}{0.066} \approx 0.030.$$

The J-cost reduction is modest ($\sim 3\%$) because stable Xe-131 still has $Z \neq N$. For comparison, transmutation to the doubly-magic nucleus $^{132}_{50}$Sn ($J_{\text{nuc}} \approx 0.125$ for $Z = 50, N = 82$) has *higher* $J(Z/N)$ than I-129 itself (because 50/82 is more asymmetric than 53/76), so it is not a J-cost-descent target in the Z/N sense. The nearest practical target is ^{136}Ba via I $\to$ Cs chains with efficiency $\eta_T \lesssim 0.10$.

Remark 5.1. The J-cost efficiency formula measures the reduction in the Z/N imbalance metric, not the actual nuclear binding energy released per transmutation. A large η_T in the J-cost sense corresponds to moving toward $Z = N$ (not necessarily toward a stable nucleus), while practical transmutation efficiency measures hazard reduction. The two do not coincide in general: ^{131}Xe is stable despite having modest J_{nuc}, while some nuclei with small J_{nuc} may still be long-lived. The RS J-cost formula provides a *structural* measure of nuclear imbalance, not a replacement for nuclear data tables.

5.3 ^{135}Cs Transmutation

^{135}Cs ($Z = 55, N = 80$): $J_{\text{nuc}} \approx 0.071$

Optimal path: Via $^{136}\text{Cs} \to {}^{136}\text{Ba}$ (stable). Only 1 neutron capture needed. Efficiency $\eta_T = 1 - J(56/80)/J(55/80) \approx 1 - 0.064/0.071 \approx 0.09$. The modest η_T reflects the fact that Ba-136 still has $Z \neq N$; the practical advantage is that ^{136}Ba is stable and non-radioactive, so the *hazard reduction* is near-perfect despite the small $J(Z/N)$ change.

6 Transmutation Resistance

Some isotopes resist transmutation because their J-cost landscape has no nearby downhill path:

$$R_T(X) = \frac{J_{\text{nuc}}(X)}{\min_{X'} J_{\text{nuc}}(X')} \tag{5}$$

where the minimum is over nuclei reachable in ≤ 5 capture/decay steps.

High R_T indicates transmutation-resistant isotopes that require more neutrons: ^{93}Zr ($R_T \approx 0.95$), ^{126}Sn ($R_T \approx 0.98$).

7 RS-Enhanced Accelerator-Driven Systems

7.1 Optimal Neutron Energy

The RS framework predicts the optimal neutron energy for transmutation:

$$E_n^{\text{opt}} = E_{\text{coh}} \cdot \varphi^r \tag{6}$$

where r is the rung separation between the fission product and its target attractor.

For most LLFPs, $r \approx 5$–10, giving $E_n^{\text{opt}} \approx E_{\text{coh}} \cdot \varphi^5$–$\varphi^{10} \approx 1$–$11$ eV (epithermal range; thermal neutrons are $\lesssim 0.025$ eV).

Remark 7.1 (Status of the optimal neutron energy formula). The formula $E_n^{\text{opt}} = E_{\text{coh}} \cdot \varphi^r$ is a *structural estimate*: neutron resonances arise from compound nucleus formation, and their positions depend on the nuclear level density and spin-orbit coupling, not directly on the φ-ladder rung of the J-cost descent path. The RS claim is that resonance energies cluster near φ-rungs (as with all energy scales in RS), and that targeting the epithermal resonance near $E_{\text{coh}} \cdot \varphi^r$ maximizes the capture cross-section. A derivation from the RS nuclear Hamiltonian is an identified open problem.

7.2 Transmutation Rate Enhancement

In RS, the transmutation rate is conjectured to be enhanced when the neutron energy matches the φ-rung of the target transition. The structural form is:

$$\sigma_{\text{RS}}(E_n) = \sigma_0 \cdot \exp\bigl(-J(E_n/E_n^{\text{opt}})\bigr)\,. \tag{7}$$

Maximum cross-section occurs at $E_n = E_n^{\text{opt}}$, since $J(1) = 0$:

$$\sigma_{\max} = \sigma_0 \cdot \exp(-J(1)) = \sigma_0 \cdot e^0 = \sigma_0. \tag{8}$$

Remark 7.2 (Status of the cross-section formula). The formula $\sigma_{\text{RS}}(E_n) = \sigma_0 \exp(-J(E_n/E_n^{\text{opt}}))$ is a *structural ansatz*, not a derivation from the RS Hamiltonian. It is motivated by the general RS principle that transitions cost J-units of recognition, and that the cost is minimized at the resonance energy E_n^{opt}. Neutron capture cross-sections in practice are determined by the Breit-Wigner resonance formula (with parameters Γ_n, Γ_γ, E_r from nuclear data) and do not take the form $\sigma_0 e^{-J(\cdot)}$ without a derivation connecting RS to these parameters. This formula should be read as a schematic RS prediction, pending a first-principles computation.

8 Conclusion

Recognition Science provides a structural framework for organizing nuclear waste transmutation. The key insights:

1. Fission products sit at J-cost local maxima on the nuclear chart
2. Doubly-magic nuclei are J-cost attractors toward which transmutation flows
3. Optimal transmutation paths follow steepest J-cost descent
4. Transmutation efficiency $\eta_T = 1 - J_f/J_0$ is maximized by targeting doubly-magic intermediaries
5. Optimal neutron energy is $E_n^{\text{opt}} = E_{\text{coh}} \cdot \varphi^r$ for rung r

This framework predicts which fission products are easy vs. resistant to transmutation (based on J-cost landscape topology) and provides design principles for accelerator-driven transmutation systems.

References

[1] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, New York (1966).

[2] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026). doi:10.3390/axioms15020090

[3] C. Rubbia *et al.*, "Conceptual Design of a Fast Neutron Operated High Power Energy Amplifier," CERN/AT/95-44 (ET), 1995.

[4] M. Salvatores *et al.*, "Transmutation of long-lived fission products in sub-critical systems," *Nuclear Science and Engineering* **145**, 63 (2003).

[5] F. G. Kondev *et al.* (Nuclear Data Sheets Editorial Board), "The NUBASE2020 evaluation of nuclear physics properties," *Chinese Physics C* **45**, 030001 (2021).